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ASCEND-RA Clinical Study Protocol



# CLINICAL STUDY PROTOCOL

## A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of MDN-4102 in Adult Subjects with Moderate-to-Severe Active Rheumatoid Arthritis

|                               |                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------|
|                               |                                                                                                        |
| Protocol Number               | MDN-4102-301                                                                                           |
| Investigational Product       | MDN-4102 (subcutaneous injection)                                                                      |
| IND Number                    | IND 000000 (fictional — placeholder)                                                                   |
| EudraCT Number                | 2026-999999-99 (fictional — placeholder)                                                               |
| ClinicalTrials.gov Identifier | NCT000000000 (fictional — placeholder)                                                                 |
| Sponsor                       | Meridian Biosciences, Inc. 1200 Innovation Parkway, Cambridge, MA 02142, USA                           |
| Sponsor Contact               | Dr Elena Whitfield, MD, PhD — VP, Clinical Development<br>medmonitor-ascendra@meridianbio-demo.example |
| Development Phase             | Phase 3                                                                                                |
| Protocol Version              | Version 2.0 (Amendment 1)                                                                              |
| Version Date                  | 15-Jun-2026                                                                                            |



# Protocol Amendment History

| Version | Date        | Type        | Summary of Changes                                                                                                                                                                                                                                                                                                   |
|---------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 01-Feb-2026 | Original    | Original approved protocol.                                                                                                                                                                                                                                                                                          |
| 2.0     | 15-Jun-2026 | Amendment 1 | Clarified Exclusion Criterion 7 (prior biologic washout); added the 24-week Open-Label Extension Period; increased planned sample size from 360 to 450 subjects to preserve power after a revised dropout assumption; added protocol-specific training requirement for site staff prior to first subject enrollment. |

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# 1. Protocol Synopsis

| <b>Title</b>                        | A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of MDN-4102 in Adult Subjects with Moderate-to-Severe Active Rheumatoid Arthritis             |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Short Title</b>                  | ASCEND-RA                                                                                                                                                                                                    |
| <b>Sponsor</b>                      | Meridian Biosciences, Inc.                                                                                                                                                                                   |
| <b>Investigational Product</b>      | MDN-4102, a humanized monoclonal antibody targeting the interleukin-6 (IL-6) receptor, administered by subcutaneous injection                                                                                |
| <b>Phase</b>                        | 3                                                                                                                                                                                                            |
| <b>Study Design</b>                 | Randomized (1:1:1), double-blind, placebo-controlled, multicenter, parallel-group study, followed by an open-label extension                                                                                 |
| <b>Study Population</b>             | Adults aged 18 to 75 years with moderate-to-severe active rheumatoid arthritis with an inadequate response to methotrexate                                                                                   |
| <b>Planned Sample Size</b>          | 450 subjects (150 per arm) at approximately 60 sites in 8 countries                                                                                                                                          |
| <b>Treatment Arms</b>               | Arm A: MDN-4102 162 mg SC every 2 weeks<br>Arm B: MDN-4102 162 mg SC every week<br>Arm C: Matching placebo SC                                                                                                |
| <b>Study Duration (per subject)</b> | Screening (up to 4 weeks) + 24-week Double-Blind Treatment Period + 24-week Open-Label Extension + 8-week Safety Follow-up                                                                                   |
| <b>Primary Objective</b>            | To evaluate the efficacy of MDN-4102 compared with placebo at Week 24 as measured by ACR20 response rate                                                                                                     |
| <b>Primary Endpoint</b>             | Proportion of subjects achieving an ACR20 response at Week 24                                                                                                                                                |
| <b>Key Secondary Endpoints</b>      | Proportion of subjects achieving ACR50 and ACR70 at Week 24; proportion achieving DAS28-CRP remission (< 2.6) at Week 24; change from baseline in HAQ-DI at Week 24; safety and tolerability through Week 56 |
| <b>Statistical Methods</b>          | Primary endpoint analyzed using the Cochran-Mantel-Haenszel test stratified by region and prior biologic exposure, within the Intent-to-Treat population, with non-responder imputation for missing data     |

## 2. Background and Rationale



### 2.1 Disease Background

Rheumatoid arthritis (RA) is a chronic, systemic autoimmune disease characterized by synovial inflammation, progressive joint destruction, and functional disability. Despite the availability of conventional synthetic and biologic disease-modifying antirheumatic drugs (DMARDs), a substantial proportion of patients fail to achieve an adequate clinical response or lose response over time, representing a persistent unmet medical need.

### 2.2 Investigational Product

MDN-4102 is a humanized IgG1 monoclonal antibody that binds with high affinity to the interleukin-6 (IL-6) receptor, blocking IL-6-mediated signaling implicated in the inflammatory cascade of RA. In a completed fictional Phase 2 study (Protocol MDN-4102-201, data on file), MDN-4102 demonstrated dose-dependent improvement in ACR20 response rates compared with placebo at Week 12, with a safety profile consistent with other agents in this class.

### 2.3 Study Rationale

This Phase 3 study is designed to confirm the efficacy and characterize the safety of two dose regimens of MDN-4102 relative to placebo in a larger, more representative population of subjects with an inadequate response to methotrexate, and to support registration of MDN-4102 for this indication.

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## 3. Study Objectives and Endpoints

### 3.1 Primary Objective and Endpoint

- Objective: To evaluate the efficacy of MDN-4102 compared with placebo at Week 24.
- Endpoint: Proportion of subjects achieving an ACR20 response at Week 24.

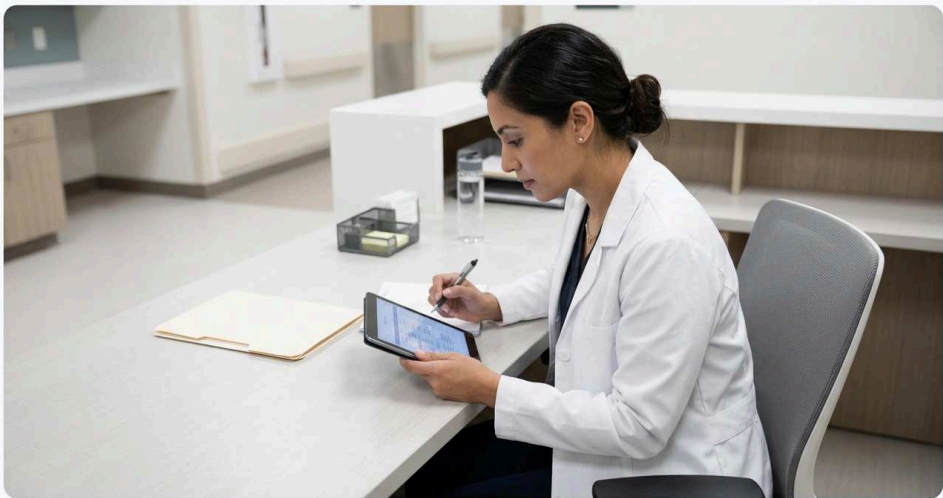
### 3.2 Secondary Objectives and Endpoints

- Proportion of subjects achieving ACR50 and ACR70 response at Week 24.
- Proportion of subjects achieving DAS28-CRP remission (score < 2.6) at Week 24.
- Change from baseline in Health Assessment Questionnaire-Disability Index (HAQ-DI) at Week 24.
- Incidence and severity of treatment-emergent adverse events (TEAEs) through Week 56.

### 3.3 Exploratory Objectives and Endpoints

- Change from baseline in select inflammatory biomarkers (CRP, IL-6, IL-6R).
- Immunogenicity, as measured by incidence of anti-drug antibodies.

## 4. Study Design



### 4.1 Overall Design

This is a randomized, double-blind, placebo-controlled, multicenter, parallel-group study consisting of a Screening Period (up to 4 weeks), a 24-week Double-Blind Treatment Period, a 24-week Open-Label Extension Period, and an 8-week Safety Follow-up Period. Eligible subjects will be randomized in a 1:1:1 ratio to receive MDN-4102 162 mg SC every 2 weeks, MDN-4102 162 mg SC every week, or matching placebo , stratified by region and prior biologic exposure.

### 4.2 Randomization and Blinding

Randomization will be performed using an interactive response technology (IRT) system. Subjects, investigators, site staff, and the sponsor's study team (with the exception of an unblinded pharmacist at each site) will remain blinded to treatment assignment throughout the Double-Blind Treatment Period. Subjects who have not achieved an ACR20 response by Week 16 will be eligible for early escape to open-label MDN-4102.

### 4.3 Schedule of Assessments

| Assessment                        | Screening | BL/D1 | Wk2 | Wk4 | Wk8 | Wk12 | Wk16 | Wk20 | Wk24 | Wk36 | Wk48 | F/U |
|-----------------------------------|-----------|-------|-----|-----|-----|------|------|------|------|------|------|-----|
| <b>Informed Consent</b>           | X         |       |     |     |     |      |      |      |      |      |      |     |
| <b>Inclusion/Exclusion Review</b> | X         | X     |     |     |     |      |      |      |      |      |      |     |
| <b>Medical History</b>            | X         |       |     |     |     |      |      |      |      |      |      |     |
| <b>Physical Examination</b>       | X         | X     |     | X   |     | X    |      | X    | X    | X    | X    |     |
| <b>Vital Signs</b>                | X         | X     | X   | X   | X   | X    | X    | X    | X    | X    | X    | X   |
| <b>Clinical Laboratory Tests</b>  | X         | X     | X   | X   | X   | X    | X    | X    | X    | X    | X    | X   |
| <b>Tuberculosis Screening</b>     | X         |       |     |     |     |      |      |      |      |      |      |     |
| <b>Randomization</b>              |           | X     |     |     |     |      |      |      |      |      |      |     |
| <b>Study Drug Administration</b>  |           | X     | X   | X   | X   | X    | X    | X    | X    | X    | X    |     |
| <b>ACR Core Set Assessment</b>    |           | X     |     | X   | X   | X    | X    | X    | X    | X    | X    |     |
| <b>DAS28-CRP</b>                  |           | X     |     | X   | X   | X    | X    | X    | X    | X    | X    |     |
| <b>HAQ-DI Questionnaire</b>       |           | X     |     |     | X   |      | X    |      | X    | X    | X    |     |

| Assessment                           | Screening | BL/D1 | Wk2 | Wk4 | Wk8 | Wk12 | Wk16 | Wk20 | Wk24 | Wk36 | Wk48 | F/U |
|--------------------------------------|-----------|-------|-----|-----|-----|------|------|------|------|------|------|-----|
| <b>Adverse Event Assessment</b>      |           | X     | X   | X   | X   | X    | X    | X    | X    | X    | X    | X   |
| <b>Concomitant Medication Review</b> |           | X     | X   | X   | X   | X    | X    | X    | X    | X    | X    | X   |
| <b>Immunogenicity Sampling</b>       |           | X     |     |     | X   |      |      |      | X    |      | X    |     |

## 5. Study Population

### 5.1 Inclusion Criteria

- Provide written informed consent prior to any study-related procedures.
- Male or female, aged 18 to 75 years, inclusive, at the time of screening.
- Diagnosis of rheumatoid arthritis per the 2010 ACR/EULAR classification criteria for at least 6 months prior to screening.
- Moderate-to-severe active disease, defined as at least 6 swollen and 6 tender joints and a high-sensitivity CRP  $\geq 1.0$  mg/dL at screening.
- Inadequate response to methotrexate at a stable dose of 15-25 mg/week for at least 12 weeks prior to randomization.
- Willing and able to comply with all study visits, procedures, and the study drug injection schedule.
- Females of childbearing potential must agree to use a highly effective method of contraception from screening through 12 weeks after the last dose of study drug.

### 5.2 Exclusion Criteria

- Prior treatment with any IL-6 or IL-6 receptor-targeted therapy.
- Prior treatment with any biologic DMARD within 3 half-lives or 8 weeks (whichever is longer) of randomization, whichever is longer, unless discussed with the Sponsor Medical Monitor.
- Active or latent tuberculosis without documented completion of appropriate prophylactic treatment.
- History of recurrent or chronic infection, or an active infection requiring systemic treatment within 4 weeks of screening.
- Diagnosis of any other inflammatory joint disease (e.g., psoriatic arthritis, ankylosing spondylitis, systemic lupus erythematosus).
- Absolute neutrophil count  $< 1,500/\text{mm}^3$ , platelet count  $< 100,000/\text{mm}^3$ , or hemoglobin  $< 8.5$  g/dL at screening.
- Alanine aminotransferase (ALT) or aspartate aminotransferase (AST)  $> 1.5\times$  the upper limit of normal at screening.
- Known history of gastrointestinal perforation or diverticulitis.
- Pregnant, breastfeeding, or planning pregnancy during the study.
- Any condition that, in the opinion of the investigator, would preclude safe participation or protocol compliance.

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## 6. Study Treatments

### 6.1 Investigational Product Description

MDN-4102 will be supplied as a sterile solution in a pre-filled syringe for subcutaneous injection, at a concentration of 162 mg/0.9 mL. Matching placebo will be identical in appearance, packaging, and labeling. Study drug must be stored refrigerated at 2-8 degrees C and protected from light.

### 6.2 Dosage and Administration

Study drug will be self-administered by subcutaneous injection at home following training by site staff at the first two dosing visits, or administered by site staff for subjects unable to self-inject. Dose interruption or discontinuation criteria are provided in Section 9 (Safety Assessments).

### 6.3 Concomitant Medications

Background methotrexate at a stable dose must be continued unchanged through Week 24. Stable doses of oral corticosteroids ( $\leq 10$  mg/day prednisone equivalent) and NSAIDs are permitted. Initiation of new DMARDs, biologics, or Janus kinase inhibitors during the Double-Blind Treatment Period is prohibited except in the case of medical necessity, and any such use must be reported to the Sponsor Medical Monitor.

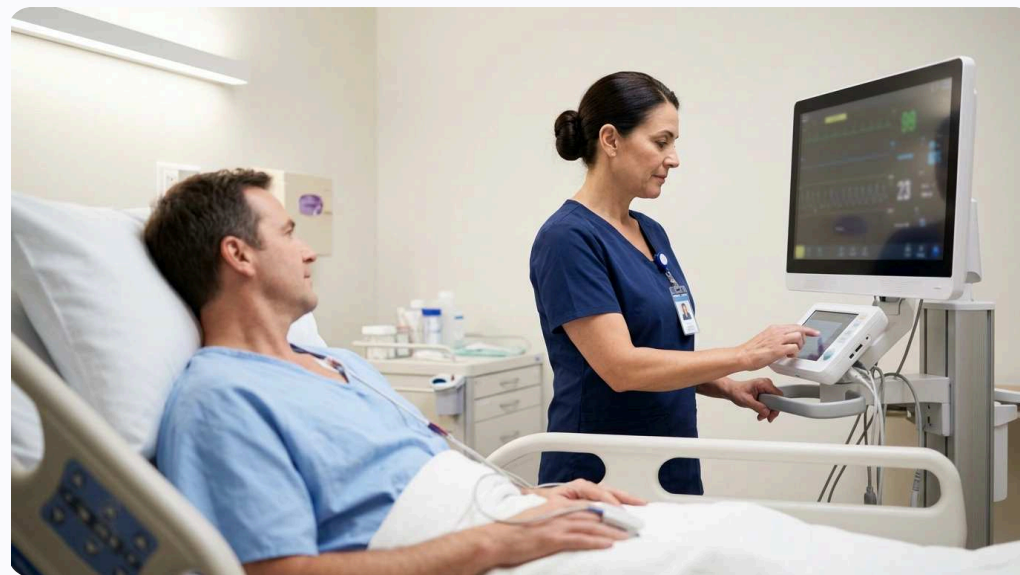
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## 7. Efficacy Assessments



Efficacy will be assessed using the ACR Core Set (tender/swollen joint counts, patient and physician global assessments, pain, HAQ-DI, and CRP), from which ACR20/50/70 response rates will be derived. Disease activity will additionally be characterized using DAS28-CRP. Assessments will be performed by a trained, blinded joint assessor at each visit specified in the Schedule of Assessments.

## 8. Safety Assessments



### 8.1 Adverse Event Definitions and Reporting

Adverse events (AEs) will be collected from the time of informed consent through the final follow-up visit. Serious adverse events (SAEs) must be reported to the sponsor's pharmacovigilance department within 24 hours of the site becoming aware of the event, regardless of causality assessment. Investigators must assess causality and expectedness for all AEs and SAEs.

### 8.2 Safety Monitoring

Safety will be monitored throughout the study by clinical laboratory testing, vital signs, physical examinations, and targeted assessment for infections (including tuberculosis reactivation), hepatic events, and injection-site reactions. An independent Data Safety Monitoring Board (DSMB) will review unblinded safety data at pre-specified intervals.

## 9. Statistical Considerations

### 9.1 Sample Size Justification

Assuming an ACR20 response rate of 60% for MDN-4102 versus 40% for placebo, a sample size of 150 subjects per arm provides at least 90% power to detect this difference at a two-sided alpha of 0.05, accounting for an estimated 15% early discontinuation rate.

### 9.2 Analysis Populations

- Intent-to-Treat (ITT) Population: All randomized subjects, analyzed according to assigned treatment; used for the primary efficacy analysis.
- Safety Population: All subjects who receive at least one dose of study drug, analyzed according to treatment actually received.
- Per-Protocol (PP) Population: ITT subjects without major protocol deviations affecting the primary endpoint; used for sensitivity analysis.

### 9.3 Primary Analysis

The primary endpoint will be analyzed using the Cochran-Mantel-Haenszel test stratified by randomization factors, within the ITT population. Subjects with missing Week 24 response status will be classified as non-responders (non-responder imputation).

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## 10. Ethical and Regulatory Considerations

This study will be conducted in accordance with the ethical principles originating from the Declaration of Helsinki, ICH E6(R2) Good Clinical Practice (GCP), and applicable local regulatory requirements. The protocol, informed consent form, and any subsequent amendments must be approved by the applicable Institutional Review Board (IRB) or Independent Ethics Committee (IEC) prior to initiation at each site. Written informed consent must be obtained from each subject, or their legally authorized representative, before performance of any study-related procedure.

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## 11. Data Management and Quality Assurance

Study data will be collected using a validated electronic data capture (EDC) system. Source data verification will be performed by clinical research associates during periodic on-site and remote monitoring visits, in accordance with the study's Monitoring Plan. The sponsor's Quality Assurance group may conduct audits of study sites, vendors, and trial master file documentation.

## 12. Study Administration



### 12.1 Protocol Amendments

Any change to this protocol will be documented in a formal amendment, reviewed and approved by the sponsor, and submitted to the applicable IRB/IEC and regulatory authorities as required before implementation, except where necessary to eliminate an immediate hazard to subjects.

### 12.2 Site Staff Training Requirement

Prior to enrolling the first subject, all site personnel performing study-related duties must complete protocol-specific training covering the study design, informed consent process, inclusion/exclusion criteria, investigational product handling and administration, and adverse event reporting procedures. Completion of this training must be documented for each individual in the site Training Log, together with role-specific delegation of duties recorded in the Delegation of Authority Log, prior to that individual performing any delegated study task. Refresher training is required following any protocol amendment that affects study conduct.

### 12.3 Record Retention

Essential documents, including signed informed consent forms, source documents, and the site Training Log, must be retained for a minimum of 25 years following study completion, or longer if required by local regulation, whichever is longer.

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## 13. References

- Aletaha D, Neogi T, Silman AJ, et al. 2010 Rheumatoid arthritis classification criteria: an American College of Rheumatology/European League Against Rheumatism collaborative initiative. *Arthritis Rheum.* 2010;62(9):2569-2581.
- International Council for Harmonisation. ICH E6(R2): Guideline for Good Clinical Practice. 2016.
- World Medical Association. Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Subjects. 2013 (as amended).
- Fries JF, Spitz P, Kraines RG, Holman HR. Measurement of patient outcome in arthritis. *Arthritis Rheum.* 1980;23(2):137-145.
- [Fictional] Meridian Biosciences, Inc. Clinical Study Report MDN-4102-201 (Phase 2). Data on file. 2025.